

Jeevan Raj M

Applied ML / LLM Engineer • Python, PyTorch, RAG & retrieval, Hugging Face

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Profile

B.E. graduate (2026) in Artificial Intelligence & Data Science (CGPA 9) with internship experience shipping Hugging Face models into a production FastAPI backend. I build applied ML and retrieval systems and judge them on held-out data: a local RAG assistant with a retrieval evaluation, a PyTorch segmentation model at 70.7% test mIoU, and a clinical NLP extractor at 0.97 F1 on held-out notes. I focus on finding where a model fails, not just where it works.

Skills

Programming: Python, SQL, TypeScript

ML / Deep Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers (Whisper, BERT, DistilBART), YOLO, ResNet, U-Net, OpenCV, model evaluation and error analysis

LLMs / Retrieval: RAG, embeddings, FAISS, chunking, retrieval evaluation (recall@k, MRR), prompt engineering, Ollama, LLM APIs (Gemini, OpenRouter)

Serving & Infra: FastAPI, REST APIs, Docker, Celery, Redis, OAuth2 / JWT / RBAC, GitHub Actions, Streamlit, AWS, Azure

Data: PostgreSQL, MySQL, SQLite, pandas, NumPy, Power BI

Experience

AI Development Intern • Kinetrix Technologies LLP

Oct 2025 – May 2026

- Built and integrated Hugging Face-based AI modules into CARE, an open-source healthcare platform: a voice-to-text clinical scribe (Whisper, BERT) and a radiology image-analysis tool (YOLO, ResNet), shipping model output into a production Python / FastAPI backend.
- Secured inference endpoints with OAuth2, JWT and RBAC; containerized both modules with Docker for AWS / Azure.
- Moved inference onto Celery workers to keep slow model calls off the request path; prepared clinical data for HL7 FHIR / DICOM.

Data Science & ML Intern • Infotact Solutions (Virtual)

Sep 2025 – Nov 2025

- Ran complete ML workflows on real-world datasets (cleaning, feature engineering, training, evaluation) with Python, scikit-learn, pandas and NumPy, choosing models on held-out metrics.

Projects

Disaster Segmentation • PyTorch, U-Net, ResNet34, OpenCV

github.com/jeevanraj-28/Disaster-segmentation

- Trained a U-Net with an ImageNet-pretrained ResNet34 encoder on FloodNet drone imagery (1,445 / 450 / 448 images, 10 classes) with cross-entropy + Dice loss, AdamW, cosine annealing and early stopping: **70.7% mean IoU** on the held-out test set.
- Per-class error analysis: vehicles and pools (under 0.2% of pixels) score lowest; flooded roads and buildings are over-predicted near water. Shipped tested train / evaluate / predict scripts.

Lumina RAG • Python, FastAPI, FAISS, Ollama, Docker

github.com/jeevanraj-28/lumina-rag-assistant

- Fully local RAG assistant: PDF / DOCX / OCR ingestion, chunking, embeddings, FAISS search and cited answers from local LLMs, with an OpenAI-compatible API.
- Built a retrieval evaluation (recall@k, MRR across chunk sizes and overlaps); a coverage test showed the chunker silently dropped **about 13% of document text**. Fixed it with regression tests, and closed an open-CORS exposure on the local API.

Clinical NLP Demo • Python, Streamlit, Hugging Face (DistilBART)

github.com/jeevanraj-28/Clinical-NLP-Demo

- Extracts complaint, diagnosis, medications (name, dose, status) and follow-up from synthetic dictation, with negation handling and a check that summaries invent no numbers.
- On 20 held-out labelled notes, raised micro F1 from **0.71 to 0.97** over the first version; negations misfiled as complaints fell from 6 of 9 to 0.

CafeCritic • Python, scikit-learn, TF-IDF, Streamlit

github.com/jeevanraj-28/cafecritic-recommender

- A data audit (474 reviews, 180 cafes, 99.4% sparse) showed every reviewer had exactly one rating, so collaborative filtering could not learn taste; redesigned as TF-IDF cafe-profile similarity plus rating, and fixed self-recommendations.

Education

B.E., Artificial Intelligence & Data Science • Mysore University School of Engineering

July 2022 – July 2026

CGPA 9. Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, DBMS, Data Structures & Algorithms.

Advanced Diploma in Computer Programming (ADMCP) • AIMIT Institute, Mysore

2024 – 2026